

Manav Batavia

Curriculum Vitae

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EDUCATION

Purdue University

Indiana, USA

Candidate for Ph.D. in Mathematics

Aug 2022 - Present

- Current GPA: 4.00/4.00.
- Advisors: Prof. Linquan Ma and Prof. Uli Walther.
- Ross-Lynn Research Fellowship, Spring 2027.
- Shreeram Abhyankar Award for Commutative Algebra, Apr 2026.
- Certificate of Merit for outstanding classroom instruction, Apr 2026.

Indian Institute of Technology Bombay

Mumbai, India

Integrated Master's of Science (B.S.+M.Sc.) in Mathematics

Jul 2017 – Jun 2022

- Awarded the Institute Silver medal.
- Cumulative Performance Index (CPI) : 9.82/10.
- Minor in Computer Science and Engineering, Minor CPI: 9.60/10.

PUBLICATIONS AND PREPRINTS

7. M. Batavia, O. Javadekar, M. Kumar, *The Betti numbers and Golodness of numerical semigroup rings of Sally type*, submitted for publication. [[arXiv](#)]
6. M. Batavia, A. Maithani, K. Mohana Sundaram, *The arithmetic rank of nullcones of classical invariant rings*, submitted for publication. [[arxiv](#)]
5. M. Batavia, C. Chen, A. N. Chlopecki, T. Duff, W. Huang, A. Li, W. Shen, *Elimination templates in Macaulay2*, to appear in Journal of Software for Algebra and Geometry. [[arXiv](#)]
4. M. Batavia, *Vanishing of local cohomology in unramified mixed characteristic*, submitted for publication. [[arXiv](#)]
3. M. Batavia, K. Mohana Sundaram, T. Murray, V. Pandey, *The arithmetic rank of residual intersections of a complete intersection*, International Mathematics Research Notices, 2026. [[doi](#)] [[arXiv](#)]
2. H. Ananthnarayan, M. Batavia, O. Javadekar, *Syzygies of associated graded modules*, to appear in Illinois Journal of Mathematics. [[arXiv](#)]
1. S. Banerjee, M. Batavia, B. Kane, M. Kyranbay, D. Park, S. Saha, H. C. So, P. Varyani, *Fermat's polygonal number theorem for repeated generalized polygonal numbers*, Journal of Number Theory, 2021. [[doi](#)] [[arXiv](#)]

INVITED TALKS

On the arithmetic rank and local cohomology of some residual intersections

- (*Upcoming*) Commutative Algebra Seminar, Purdue University, Oct 2026.

A sharp upper bound on cohomological dimension in unramified mixed characteristic

- (*Upcoming*) Joint Mathematics Meeting, Jan 2027.
- (*Upcoming*) Algebraic Geometry Seminar, University of Illinois Chicago, Nov 2026.
- (*Upcoming*) Algebraic Geometry and Commutative Algebra Seminar, University of Notre Dame, Nov 2026.
- (*Upcoming*) Commutative Algebra Seminar, University of Nebraska-Lincoln, Nov 2026.
- (*Upcoming*) Poster at KUMUNU, University of Kansas, Oct 2026
- Algebra Seminar, University of Missouri, Sep 2026.
- Poster at Pan-American School in Commutative Algebra 2.0, Centro de Investigación en Matemáticas, May 2026.

- Commutative Algebra and Algebraic Geometry Seminar, University of Minnesota, Mar 2026.
- Commutative Algebra Seminar, Purdue University, Feb 2026.

Elimination Templates in Macaulay2

- SIAM Annual Meeting, Cleveland, Jul 2026.

The arithmetic rank of residual intersections of a complete intersection

- AMS Spring Central Sectional Meeting, North Dakota State University, Apr 2026.
- AMS Fall Central Sectional Meeting, St. Louis University, Oct 2025.
- Commutative Algebra Seminar, University of Utah, Sep 2025.
- Commutative Algebra Seminar, Purdue University, Sep 2025.

Cohomological dimension and arithmetic rank

- Turbo Talk, CA+, Iowa State University, Mar 2026.

When do associated graded modules have pure resolution?

- Poster at Connections Workshop in Commutative Algebra, Simons Laufer Mathematical Institute, Jan 2024.

CONFERENCES/WORKSHOPS

(Upcoming) Joint Mathematics Meeting <i>Chicago</i>	Jan 2027
(Upcoming) KUMUNU <i>University of Kansas</i>	Oct 2026
Upcoming Researchers in Commutative Algebra (URiCA) <i>University of Nebraska-Lincoln</i>	Aug 2026
International Congress of Mathematics <i>Philadelphia</i>	Jul 2026
SIAM Annual Meeting <i>Cleveland</i>	Jul 2026
Commutative Algebra and Algebraic Geometry in Milan <i>Politecnico di Milano</i>	Jun 2026
Pan-American School in Commutative Algebra (PASCA) 2.0 <i>Centro de Investigación en Matemáticas</i>	May 2026
AMS Spring Central Sectional Meeting <i>North Dakota State University</i>	Apr 2026
CA+ <i>Iowa State University</i>	Mar 2026
Commutative Algebra and Modular Representation Theory <i>IIT Madras</i>	Dec 2025
AMS Fall Central Sectional Meeting <i>University of St. Louis</i>	Oct 2025
Macaulay2 Workshop <i>University of Wisconsin-Madison</i>	Jun 2025
Singularities in Algebra and Geometry <i>Centro de Investigación en Matemáticas</i>	Jun 2025
Séminaire de Mathématiques Supérieures <i>University of Toronto</i>	Jun 2025
Commutative Algebra in Mixed Characteristic <i>University of Nebraska-Lincoln</i>	Mar 2025
Howard Rowlee Lecture <i>University of Nebraska-Lincoln</i>	Mar 2025
CA+ <i>Iowa State University</i>	Mar 2025
Cohomologies of Commutative Algebras <i>IIT Madras</i>	Dec 2024
UweFest <i>University of Notre Dame</i>	Aug 2024
Workshop on Recent Trends in Commutative Algebra <i>IIT Bombay</i>	Jun 2024
International Conference on Local Rings and Singularities <i>IIT Bombay</i>	Jun 2024
Upcoming Researchers in Commutative Algebra (URiCA) <i>University of Nebraska-Lincoln</i>	May 2024
Introductory Workshop: Commutative Algebra <i>SLMath</i>	Jan 2024
Connections Workshop: Commutative Algebra <i>SLMath</i>	Jan 2024
Workshop on Commutative Algebra and Algebraic Geometry in Prime Characteristics <i>ICTP</i>	May 2023

TECHNICAL SKILLS

Programming Languages: C++, Python, MATLAB.

Softwares: SageMath, Macaulay2, L^AT_EX, SolidWorks, AutoCAD, XCircuit.

ACADEMIC ACHIEVEMENTS

- Recipient of the Shreeram Abhyankar Award for Commutative Algebra, awarded by the Department of Mathematics at Purdue University, 2026.
- Department Rank 1 in mathematics at IIT Bombay, 2022.
- Awarded the Mitacs Globalink scholarship for a research internship in Canada, 2021.
- All India Rank of 174 in IIT JEE Advanced (out of more than 200,000 candidates), 2017.
- All India Rank of 8 in the B.Math entrance exam for Indian Statistical Institute (ISI), 2017.

UNDERGRADUATE RESEARCH EXPERIENCE

Master's Thesis and Further Research

Jul 2021 – Dec 2022

Guide: Prof. Ananthnarayan Hariharan, Department of Mathematics

IIT Bombay

- Proved a necessary and sufficient condition for the existence of a pure resolution of the associated graded module corresponding to a module over a Noetherian local ring.
- Independently proved the Auslander-Buchsbaum-Serre equivalences, and studied Burch's work on constructing ideals with arbitrarily large projective dimension, generated by three or fewer elements.

Betti Tables of Quadratic Ideals in Characteristic Two

Jun - Aug 2021

Guide: Prof. Alexandra Seceleanu, Department of Mathematics

University of Nebraska-Lincoln

This research was done as part of the Polymath Jr. 2021 program.

- Succeeded in resolving all quadratic ideals with three generators in a polynomial ring in three variables over a field of characteristic two. The results attained are easily applicable to larger ideals and larger polynomial rings.
- Presented our results at the Young Mathematicians Conference 2021, organised by The Ohio State University.

Generalization of Fermat's Polygonal Number Theorem

May – Jul 2019

Guide: Prof. Ben Kane, Department of Mathematics

The University of Hong Kong

- Acquired minimal number of terms for which certain weighted sums of generalized polygonal numbers are universal.
- The work done resulted in the aforementioned publication in the Journal of Number Theory.

TEACHING EXPERIENCE

Recipient of the 2026 Certificate of Merit at Purdue University for outstanding classroom instruction.

Instructor of Record | *Purdue University*

- Algebra Qualifying Exam Prep Class, Summer 2025
- Applied Calculus I, Fall 2024

Recitation/Tutorial Instructor

- Calculus III | Fall 2023, Spring 2024, Summer 2024 and Spring 2026, Purdue University
- Calculus II | Fall 2022 and Spring 2023, Purdue University
- Linear Algebra | Spring 2022, IIT Bombay
- Calculus I | Fall 2019, Fall 2022, IIT Bombay

SERVICE

- Organizer of Ideal Conversations, the student commutative algebra seminar at Purdue University. (Aug 2024 - Present)
- Association for Women in Mathematics (AWM) mentor for graduate students at Purdue University. (Aug 2023 - Present)
- Founder of the Department Academic Mentorship Programme in the Department of Mathematics at IIT Bombay.
- Department Academic Mentor for undergraduate math students at IIT Bombay. (Apr 2019 - May 2022)
- Volunteer for the National Service Scheme (NSS) at IIT Bombay, creating waste awareness and imparting environmental education to underprivileged children. (Aug 2017 - May 2018)